ENGINEERING ECONOMY

LECTURE 7

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Annual Worth Analysis

- We previously learned the present worth (PW) method and its companion, the future worth (FW) method.
- Here we learn the equivalent annual worth, or AW, method. AW analysis is commonly considered the more desirable of the two methods because the AW value is easy to calculate;
- the measure of worth—AW in monetary units per year—is understood by most individuals,
- and its assumptions are essentially identical to those of the PW method.
- Annual worth is also known by other titles. Some are equivalent annual worth (EAW), equivalent annual cost (EAC), annual equivalent (AE), and equivalent uniform annual cost (EUAC).

- ***AW* value is** the equivalent uniform annual worth of all estimated receipts and disbursements during the life cycle of the project or alternative.
- *AW* value is equivalent to the *PW* and *FW* values at the MARR for *n* years
- $AW = PW(A/P, i, n) = FW(A/F, i, n)$
- where *n* is the number of years for equal-service comparison.
- When all cash flow estimates are converted to an *AW* value, this value applies for:
  - every year of the life cycle
  - for each additional life cycle
- The prime advantage of *AW* analysis is:

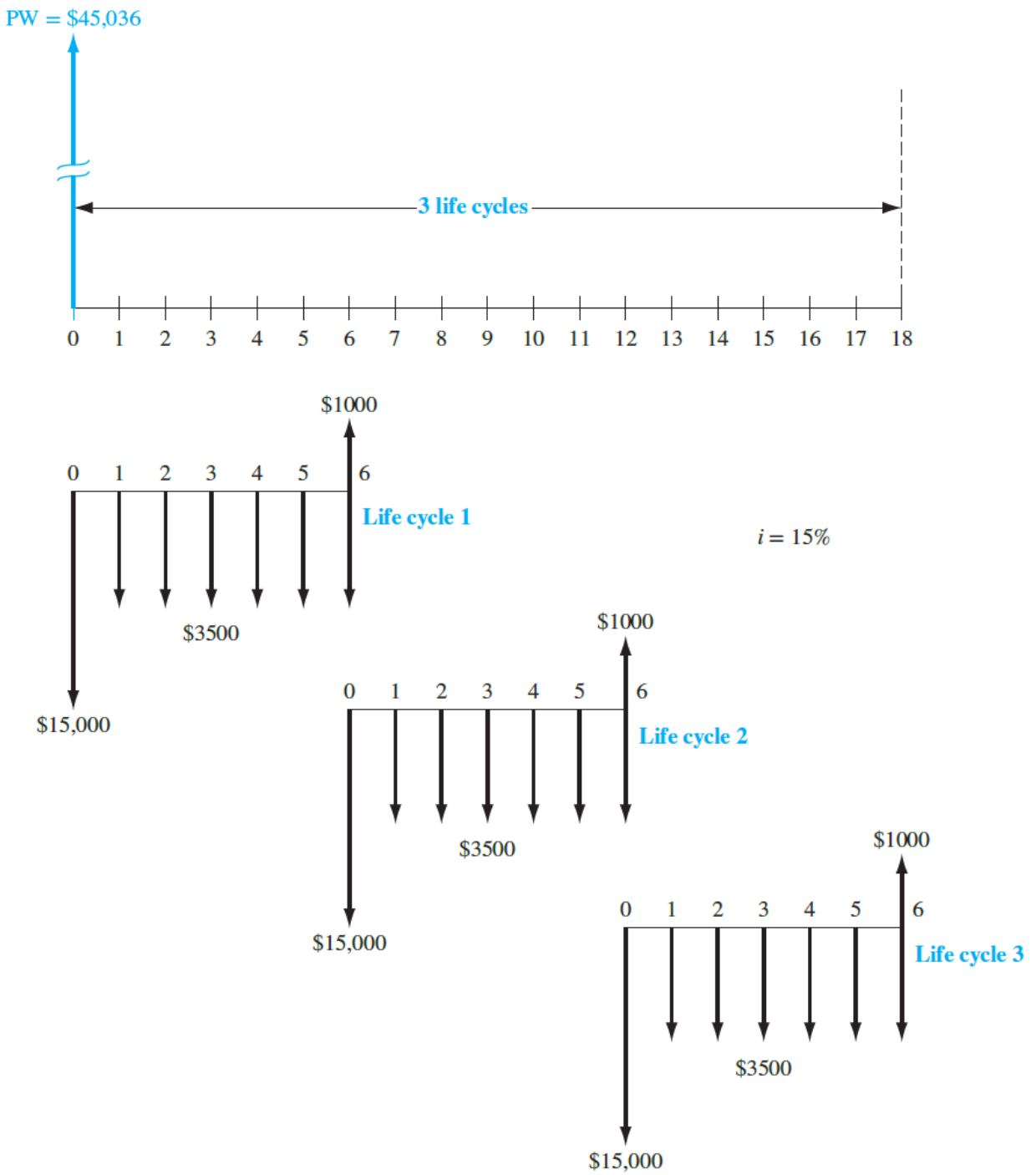
  1) AW value has to be calculated for only one life cycle

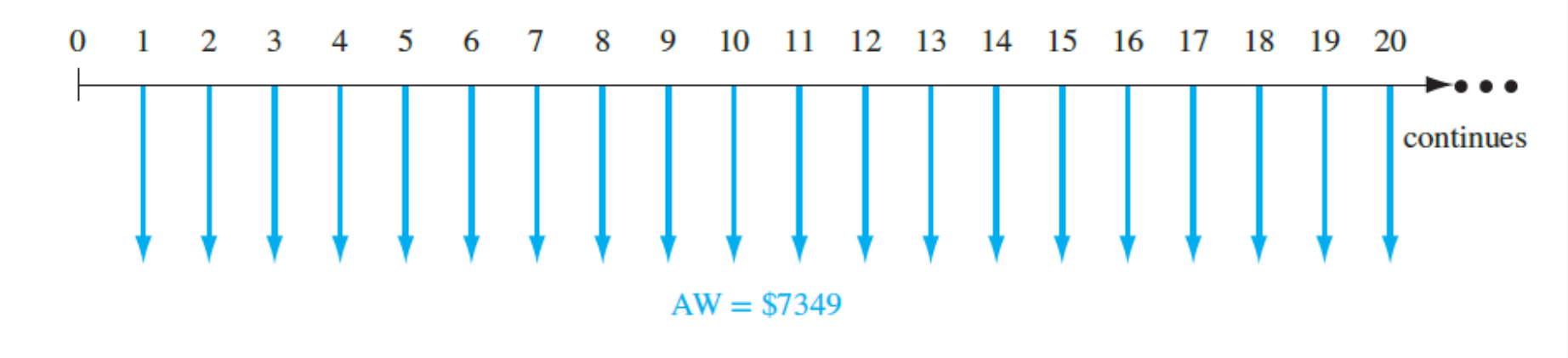
  2) not necessary to use the LCM of lives as it is for *PW* and *FW*.

- When alternatives being compared have different lives, the *AW* method makes the assumption that:
    1) The service provided by alternatives will be needed for at least the LCM of years or more
    2) The selected alternative will be repeated over each life cycle of the LCM in exactly the same manner
    3) The cash flow estimates will be the same in every life cycle

EXAMPLE

- In one of our previous Example, National Homebuilders, Inc. evaluated cut-and-finish equipment from vendor A (6-year life) and vendor B (9-year life). The PW analysis used the LCM of 18 years. Consider only the vendor A option now.
- The diagram shows the cash flows for all three life cycles (first cost $−15,000; annual M&O costs $−3500; salvage value $1000).
- Demonstrate the equivalence at i = 15% of PW over three life cycles and AW over one cycle. Previously present worth for vendor A was calculated as PW = $−45,036.


PW = $45,036
3 life cycles
0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18
$1000
0 1 2 3 4 5 6
Life cycle 1
$3500
$15,000
i = 15%
$1000
0 1 2 3 4 5 6
Life cycle 2
$3500
$15,000
$1000
0 1 2 3 4 5 6
Life cycle 3
$3500
$15,000



- Annual Worth (AW) answers the question:
- "If this project kept repeating forever, what constant amount of money per year would be financially equivalent to all its actual cash flows?"
- Calculate the equivalent uniform AW value for all cash flows in the first life cycle.

$$AW = -15{,}000(A/P,15\%,6) + 1000(A/F,15\%,6) - 3500 = \$-7349$$

- When the same computation is performed on each succeeding life cycle, the AW value is $−7349. Now the PW value for 18 years.

$$AW = -45{,}036(A/P,15\%,18) = \$-7349$$

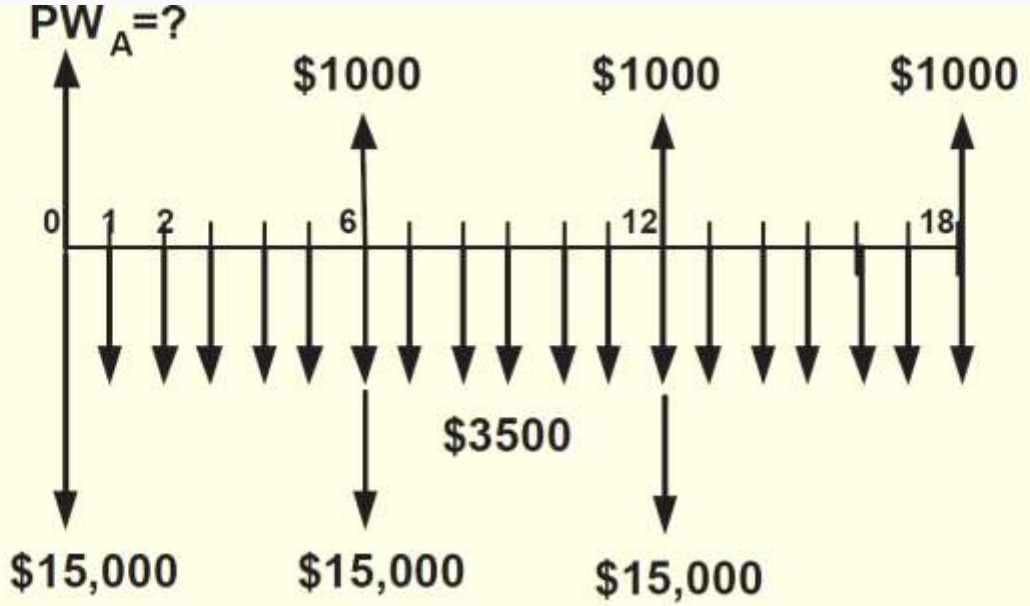
- The one-life-cycle AW value and the AW value based on 18 years are equal.

Example

Eric Forman, a project engineer is assigned to start up a new office in Wisconsin. Two lease options are available:

	Location A	Location B
First cost, $	$-15,000	$-18,000
Annual lease cost, $ per year	-3,500	-3,100
Deposit return, $	1,000	2,000
Lease term, years	6	9

For Location A, demonstrate the the equivalence at $i = 10\%$ of PW ($-55,888.4) over three life cycles and AW over one cycle.



1. From $PW = \$ - 55,888.4$ over 18 years

$$AW = PW(A/P, 10\%, 18) = -55,888.4(0.12193) = -6,814.47$$

2. Directly from AW over 6 years

$$\begin{aligned} AW &= -15,000(A/P, 10\%, 6) - 3500 + 1,000(A/F, 10\%, 6) \\ &= -15,000(0.22961) - 3500 + 1,000(0.12961) \\ &= \$ - 6,814.54 \end{aligned}$$

DIFFERENT CASES FOR AW CALCULATION

- **Example 1 — Initial Cost + Annual Operating Cost**
- **Initial cost:** $12,000
  Annual operating cost: $2,000
  Life: 5 years
  i = 10%

  $$AW = -12{,}000(A/P,10\%,5) - 2{,}000$$

- Factor:

  $$A/P = 0.2638$$

  $$AW = -12{,}000(0.2638) - 2{,}000 = -5{,}166$$

- **Annual Worth = –$5,166**

- **Example 2 — Initial Cost + Annual Cost + Salvage Value**
- **Initial cost:** $20,000
  Salvage value (Year 8): $4,000
  Annual cost: $1,500
  Life: 8 years
  i = 8%
- $AW = -20{,}000(A/P,8\%,8) - 1{,}500 + 4{,}000(A/F,8\%,8)$
- Factors:

  $A/P = 0.1740, \quad A/F = 0.0890$

  $AW = -3{,}480 - 1{,}500 + 356 = -4{,}624$

- **Annual Worth = –$4,624**

- **Example 3 — Pure Annual Revenue (Uniform Series)**
- **Annual revenue:** $7,000
  Goes on indefinitely
  i = 12%
- This is already an annual series:

  AW = +7,000AW
- **Annual Worth = +$7,000**

- **Example 4 — Single Future Benefit + Annual Operating Cost**
- **Future revenue (Year 10):** $30,000
  Annual cost: –$1,200
  i = 10%

  $AW = -1{,}200 + 30{,}000(A/F,10\%,10)$

- Factor:

  $A/F = 0.06543$

  $AW = -1{,}200 + 1{,}963 = +763$

- **Annual Worth = +$763**

- **Example 5 — One Cost at Year 0 + One Cost at Year N**
- **Cost at Year 0:** −$8,000
  Cost at Year 6: −$5,000
  i = 10%

  $AW = -8{,}000(A/P,10\%,6) - 5{,}000(A/F,10\%,6)$
- Factors:

  $A/P = 0.22961, \quad A/F = 0.12961$

  $AW = -1{,}836.9 - 648 = -2{,}484.9$
- **Annual Worth = −$2,485**

- **Example 6 — Initial Cost + Increasing (Gradient) Annual Costs**
- **Initial cost:** $10,000
  First-year cost: $500
  Annual increase: +$200 per year
  Life: 5 years
  i = 10%
- Gradient factor (table):

  A/G=1.5770

  AW=−10,000(A/P,10%,5)−500−200(1.5770)
- Factor:

  A/P=0.2638

  AW=−2,638−500−315.4=−3,453.4
- **Annual Worth = –$3,453**

- Selection guidelines are the same as for the *PW* method
- For Mutually Exclusive Alternatives:

  1) <u>One Alternative</u>: Calculate *AW* at MARR. If *AW* ≥ 0, the requested MARR is met or exceeded and the alternative is financially viable.

  2) <u>Two or more Alternatives</u>: Calculate *AW* of each alternative at MARR. Select the alternative with the *AW* value that is numerically largest.

- For independent projects:
  - All projects with *AW* ≥ 0 calculated at MARR are acceptable.
- If a study period is specified, the cash flows over the study period are converted to *AW*.

Calculation of Capital Recovery and AW Values

- an alternative should have the following cash flows estimated as accurately as possible:
- **Initial investment *P.*** This is the total first cost of all assets and services required to initiate the alternative. When portions of these investments take place over several years, their PW is an equivalent initial investment. Use this amount as *P*.
- **Salvage value *S.*** This is the terminal estimated value of assets at the end of their useful life. The *S* is zero if no salvage is anticipated; *S* is negative when it will cost money to dispose of the assets. For study periods shorter than the useful life, *S* is the estimated market value or trade-in value at the end of the study period.
- **Annual amount *A.*** This is the equivalent annual amount (costs only for cost alternatives; costs and receipts for revenue alternatives). Often this is the annual operating cost (AOC) or M&O cost, so the estimate is already an equivalent *A* value.

- The AW value for an alternative is comprised of two components: **capital recovery** for the initial investment *P* at a stated interest rate (usually the MARR) and the equivalent **annual amount *A*.** The symbol CR is used for the capital recovery component.
- In equation form,

$$AW = CR + A$$

- The recovery of an amount of capital *P* committed to an asset, **plus** the time value of the capital at a particular interest rate, is a **fundamental principle of economic analysis.**
- Capital recovery (CR) is the equivalent annual amount that the asset, process, or system must earn (new revenue) each year to just **recover the initial investment plus a stated rate of return** over its expected life. Any expected salvage value is considered in the computation of CR.

- So, **CR is the annual amount needed to recover:**
  - the initial investment PPP
  - **plus** the interest (time value of money)
  - **minus** salvage value (if any)

- Accordingly, CR is calculated as:

$$CR = -P(A/P,i,n) + S(A/F,i,n)$$

- **The Component A (Annual Operating Benefit or Cost)**
- This is:
  - operating cost (if negative),
  - revenue (if positive),
  - maintenance expense,
  - fuel, labor, etc.
- In short:
- A = all other annual amounts that occur every year.

$$AW = \underbrace{\text{amount needed to recover the investment}}_{\text{CR}} + \underbrace{\text{other annual benefits/costs}}_{A}$$

This gives the **total annual economic effect** of the alternative.

EXAMPLE

- Lockheed Martin is increasing its booster thrust power in order to win more satellite launch contracts from European companies interested in opening up new global communications markets. A piece of earth-based tracking equipment is expected to require an investment of $13 million, with $8 million committed now and the remaining $5 million expended at the end of year 1 of the project. Annual operating costs for the system are expected to start the first year and continue at $0.9 million per year. The useful life of the tracker is 8 years with a salvage value of $0.5 million. Calculate the CR and AW values for the system, if the corporate MARR is 12% per year.

- *Capital recovery:* Determine *P* in year 0 of the two initial investment amounts, calculate the CR. In $1 million units,

$$P = 8 + 5(P/F,12\%,1) = \$12.46$$

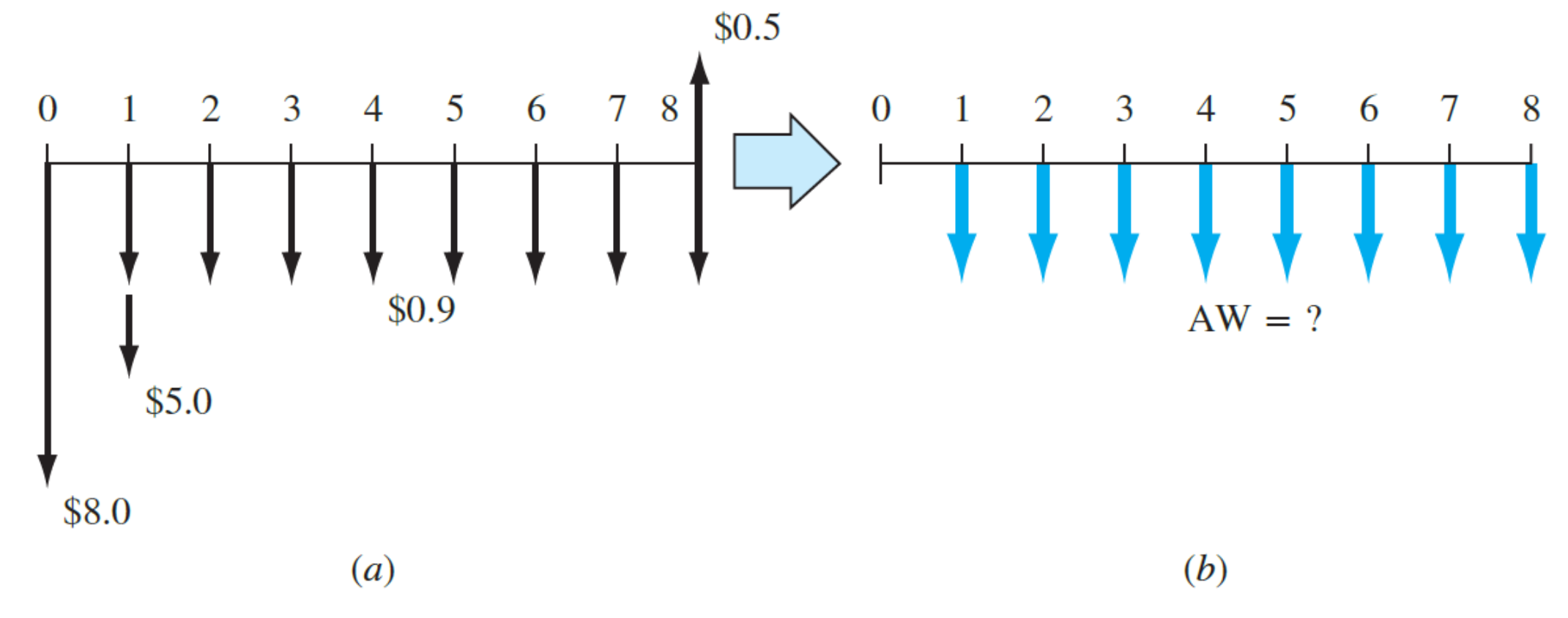
$$\text{CR} = -12.46(A/P,12\%,8) + 0.5(A/F,12\%,8)$$

$$= -12.46(0.20130) + 0.5(0.08130)$$

$$= \$-2.47$$

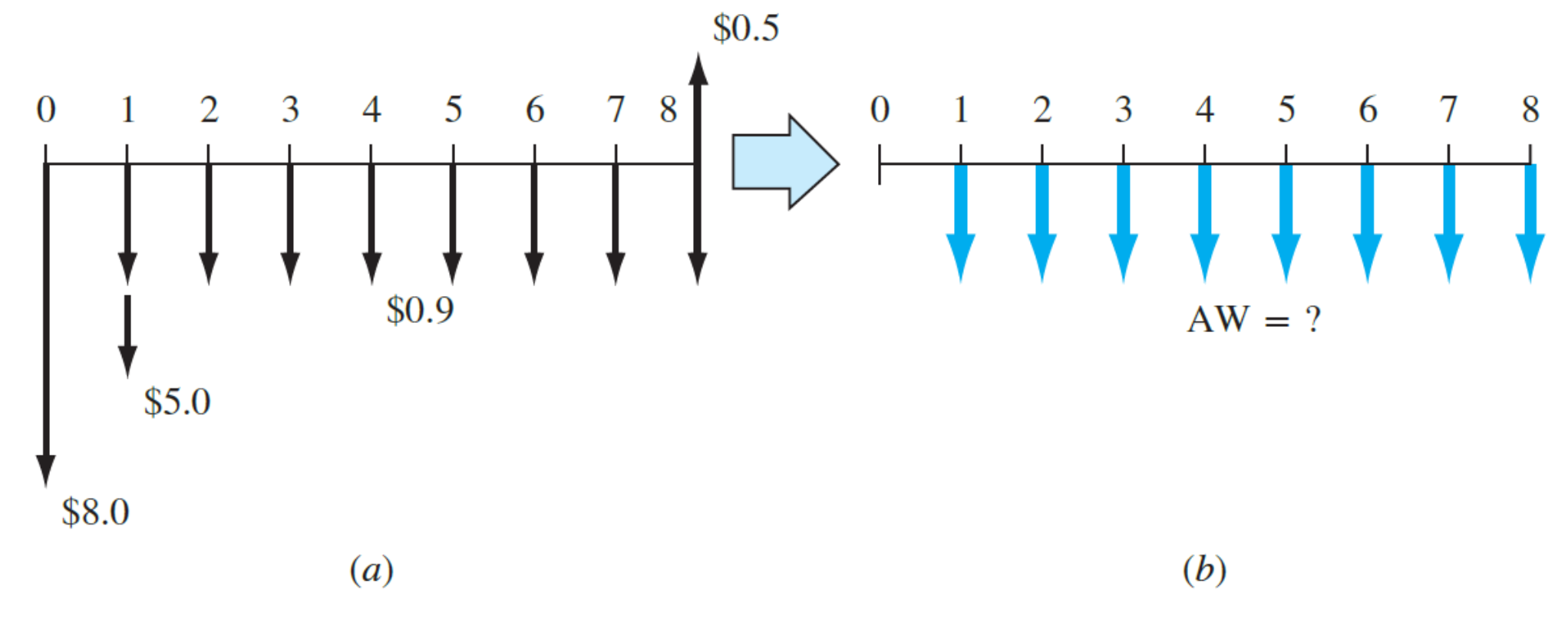
- The correct interpretation of this result is very important to Lockheed Martin.
- It means that each and every year for 8 years, the equivalent total net revenue from the tracker must be at least $2,470,000 just to recover the initial present worth investment plus the required return of 12% per year. This does not include the AOC of $0.9 million each year.

- *Annual worth:* To determine AW, the cash flows in Figure a must be converted to an equivalent AW series over 8 years (Figure b).
- Since CR = $−2.47 million is an *equivalent annual cost,* as indicated by the minus sign, total AW is easily determined by adding the CR and AOC values.



AW = −2.47 − 0.9 = $−3.37 million per year

- This is the AW for all future life cycles of 8 years, provided the costs rise at the same rate as inflation, and the same costs and services are expected to apply for each succeeding life cycle.



Southern Cement plans to open a new rock mining site. Two plans are suggested. Plan A requires the purchase of two earth movers and an unloading pad at the plant. Plan B calls for the construction of a conveyor from mining site to the plant. MARR is effective 15% per year compounded monthly, and costs are given:

	Plan A		Plan B
	Mover	Pad	Conveyor
First cost,$	$-45,000	$-28,000	-175,000
Annual Operating cost, $	-6,000	-300	-2,500
Salvage value, $	5,000	2000	10,000
Life, years	8	12	24

1. Compare the two plans using *AW* method
2. Compare the two plans using *AW* method over a study period of 6 years. The market value of each mover after 6 years is $20,000 and trade-in value of the conveyor after 6 years is $25,000. The pad can be salvaged for $2,000.

a) Earth Movers (two units)

- From the problem statement:
- Initial cost per mover = $45,000
  → Since two movers are required, the total **initial cost P = $90,000**
- Annual operating cost per mover = $6,000
  → For two movers, **annual cost A = –$12,000 per year**
- Salvage value per mover = $5,000
  → For two movers, **salvage value S = $10,000**
- Useful life = **8 years**

- The annual worth for the movers consists of:
- **Capital recovery (CR)** using

$$CR=-P(A/P,15\%,8)+S(A/F,15\%,8)$$

- **Plus the annual operating cost**
- Therefore:

$$AW_{movers}=-90{,}000(A/P,15\%,8)+10{,}000(A/F,15\%,8)-12{,}000$$

- This corresponds to the first line of the solution:

$$-12{,}000-90{,}000(A/P,15\%,8)+10{,}000(A/F,15\%,8)$$

b) Unloading Pad

- From the problem data:
- Initial cost = **P = $28,000**
- Annual operating cost = **A = –$300 per year**
- Salvage value = **S = $2,000**
- Useful life = **12 years**
- The annual worth of the pad is:

$$AW_{pad} = -28{,}000(A/P,15\%,12) + 2{,}000(A/F,15\%,12) - 300$$

- This corresponds to:

$$-300 - 28{,}000(A/P,15\%,12) + 2{,}000(A/F,15\%,12)$$

1. The LCM of lives is 24 years.

$$
\begin{aligned}
AW_A &= -12,000 - 90,000(A/P, 15\%, 8) + 10,000(A/F, 15\%, 8) \\
&- 300 - 28,000(A/P, 15\%, 12) + 2,000(A/F, 15\%, 12) \\
&= -12,000 - 90,000(0.22285) + 10,000(0.07285) \\
&- 300 - 28,000(0.18448) + 2,000(0.03448) \\
&= \$ - 36,724
\end{aligned}
$$

Given Data for Plan B

- Initial cost:

  $P = \$175{,}000$

- Annual operating cost:

  $A = -\$2{,}500$ per year

- Salvage value:

  $S = \$10{,}000$

- Useful life:

  $n = 24$ years

- MARR = 15%

$$
\begin{aligned}
AW_B &= -175,000(A/P,15\%,24) + 10,000(A/F,15\%,24) \\
&- 2,500 \\
&= -175,000(0.15543) - 2,500 + 10,000(0.00543) \\
&= \$ -29,646
\end{aligned}
$$

The Annual Worth (AW) values for the two alternatives are:

$$AW_A = -\$36{,}724$$

$$AW_B = -\$29{,}646$$

Since both alternatives represent costs, **the option with the less negative AW value is economically preferred**.

$$AW_B > AW_A \quad \Rightarrow \quad -29{,}646 > -36{,}724$$

Therefore, **Plan B is more cost-effective**, because it results in a smaller equivalent annual cost over the

1. **Plan B is selected**
2. Over a study period of 6 years

$$
\begin{aligned}
AW_A &= -12,000 - 90,000(A/P, 15\%, 6) + 40,000(A/F, 15\%, 6) \\
&- 300 - 28,000(A/P, 15\%, 6) + 2,000(A/F, 15\%, 6) \\
&= -12,000 - 90,000(0.26424) + 40,000(0.11424) \\
&- 300 - 28,000(0.26424) + 2,000(0.11424) \\
&= \$ - 38,682 \\
AW_B &= -175,000(A/P, 15\%, 6) + 25,000(A/F, 15\%, 6) \\
&- 2,500 \\
&= -175,000(0.26424) - 2,500 + 25,000(0.11424) \\
&= \$ - 45,886
\end{aligned}
$$

Plan A is selected

- Same concept in Capitalized Cost, projects that have long lives that can be considered infinite in economic analysis
- Annual worth of an initial investment is $A = P\,i$.
- Cash flows recurring regular or irregular intervals handled by:
- converting them to equivalent uniform annual amounts A for one cycle

Example

An engineer received a bonus of \$10,000. If he deposits it now at an interest rate of 8% per year, how many years must the money accumulate before she can withdraw \$2,000 per year forever?

$$\begin{aligned} P_n = \frac{A}{i} &= \frac{2,000}{0.08} = \$25,000 \\ 10,000(1+0.08)^n &= 25,000 \rightarrow n = 12 \end{aligned}$$

Rate of Return Analysis: One Project

- Along with the *PW* and *AW* criteria, the third primary measure of worth is rate of return
- Mostly preferred over *PW* and *AW* criteria. Ex: an alternative's worth can be reported as:
  - 15% rate of return on the investment (EXACT!)
  - will result in a net surplus of $10,000 in terms of *PW* at 10% MARR (return > 10% is known only)
- ROR is also known as
  - Internal Rate of Return (IRR)
  - Return on Investment (ROI)

- **The AW method is very powerful when cash flows are irregular.**
- Many engineering projects do **not** generate equal annual costs or benefits.

For example:

- Some years have **very high expenses**
- Some years have **low expenses**
- Some years have **no payments at all**
- Some years may even have **positive revenue**
- Because the yearly amounts fluctuate, the **Annual Worth (AW)** method is useful for converting all of these irregular cash flows into an equivalent uniform annual value.
- This allows for a fair year-to-year comparison.

- **However, in some cases AW is *not* the preferred method.**

Capital-intensive projects

- These are projects that require a **very large initial investment**, such as:
- Industrial plants
- Dams
- Tunnels
- Power stations
- Such projects typically involve huge upfront costs and long-term benefits.

- In these cases, it is usually easier and more appropriate to use:
- **Present Worth (PW)** or
  Internal Rate of Return (IRR) methods.

Projects with clear long-term trends

- Some projects exhibit a predictable **trend** in costs or revenues:
- Revenues increase by 5% each year
- Maintenance costs rise by 20% annually
- System capacity changes over time
- The AW method becomes less convenient when values change systematically over time.

- Better choices are:
- **PW** or long-term **NPV (Net Present Value)** analysis

Projects with intermittent (non-annual) costs

Examples include:

- Major overhaul every 5 years
- System replacement every 10 years
- Licensing fees that occur every 4 years

- When costs do *not* occur annually, forcing them into an annual equivalent may distort the analysis.

- Thus, it is usually more practical to apply:
- **PW**, which handles irregular and lump-sum cash flows directly.

- AW is most suitable for comparing alternatives on an annual basis, especially when yearly cash flows vary.
- But when a project involves very large initial investments, trends, or intermittent cost patterns, the PW or IRR methods generally provide clearer insight.

- **Rate of Return: Definition 1**
- is the rate (−100% to 1) paid on the unpaid balance of borrowed money
- The rate of return is calculated on the unrecovered balance not the initial amount of investment.

Example

Suppose a bank lends \$10,000, which is repaid in installments of \$4,021 at the end of each year for three years. What is the interest rate the bank is charging for this transaction?

$$\$10,000 = \$4,021(P/A, i, 3) \rightarrow 10\%$$

What is the meaning of 10%?

Year	Unpaid Balance at Beginning	Return on Unpaid Balance	Payment Received	Balance at End
0				-\$10,000
1	-\$10,000	-\$1,000	\$4,021	-\$6,979
2	-\$6,979	-\$698	\$4,021	-\$3,656
3	-\$3,656	-\$365	\$4,021	\$0

- Year 1
- **Beginning balance**

  $-10{,}000$
- **Interest (Return on balance)**

  $10\% \times 10{,}000 = 1{,}000$
- **Payment**

  Borrower pays:

  4,0214
- **Ending balance**

  $-10{,}000 - 1{,}000 + 4{,}021 = -6{,}979$

So after one year, the borrower still owes **$6,979.**

- Year 2
- Beginning balance

  −6,979
- Interest

  10%×6,979=698
- Payment

  4,0214
- Ending balance

  −6,979−698+4,021=−3,656

Now the borrower owes **$3,656**.

- Year 3
- Beginning balance

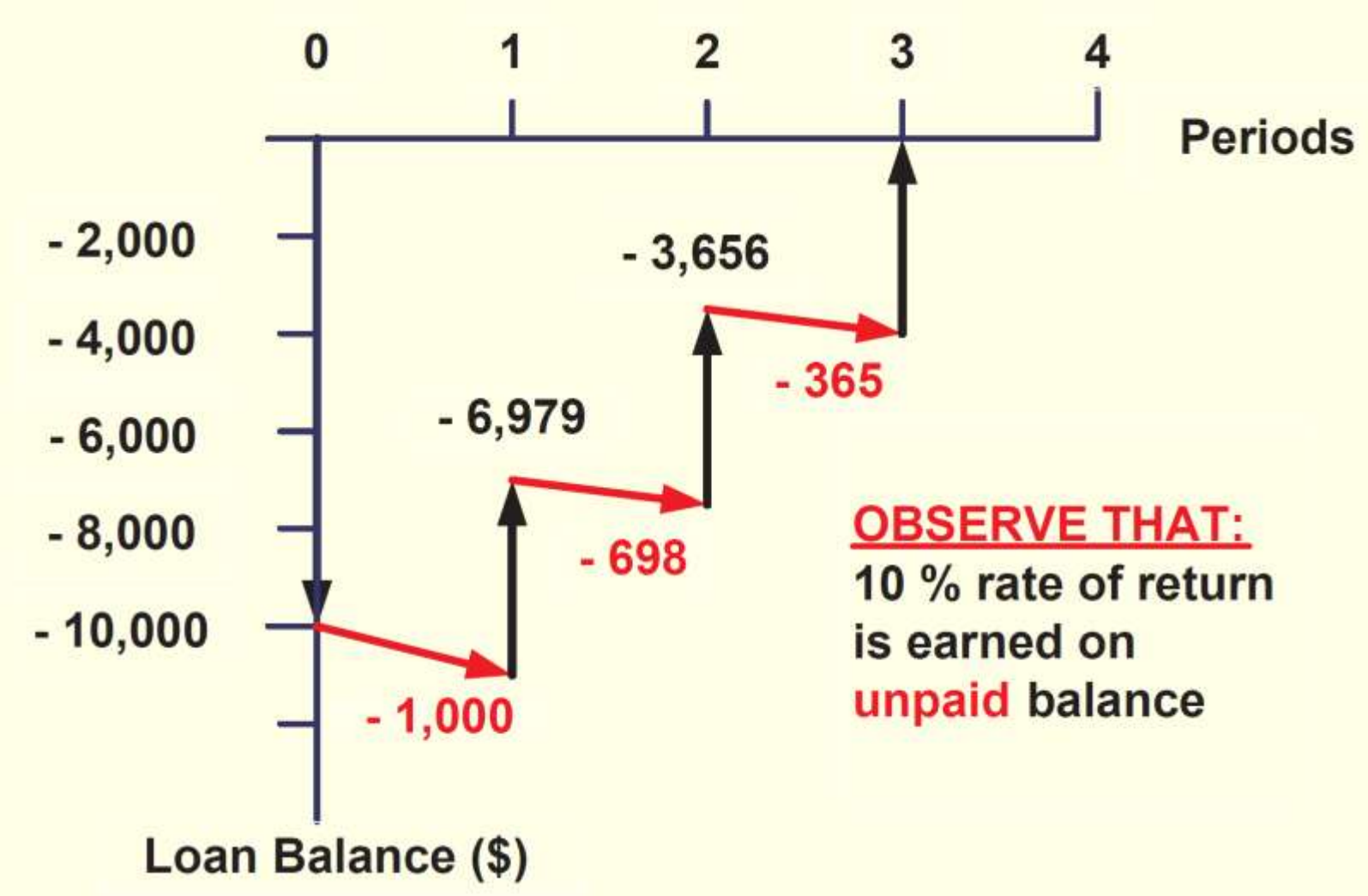
−3,656
- Interest

  10%×3,656=365.6 65
- Payment

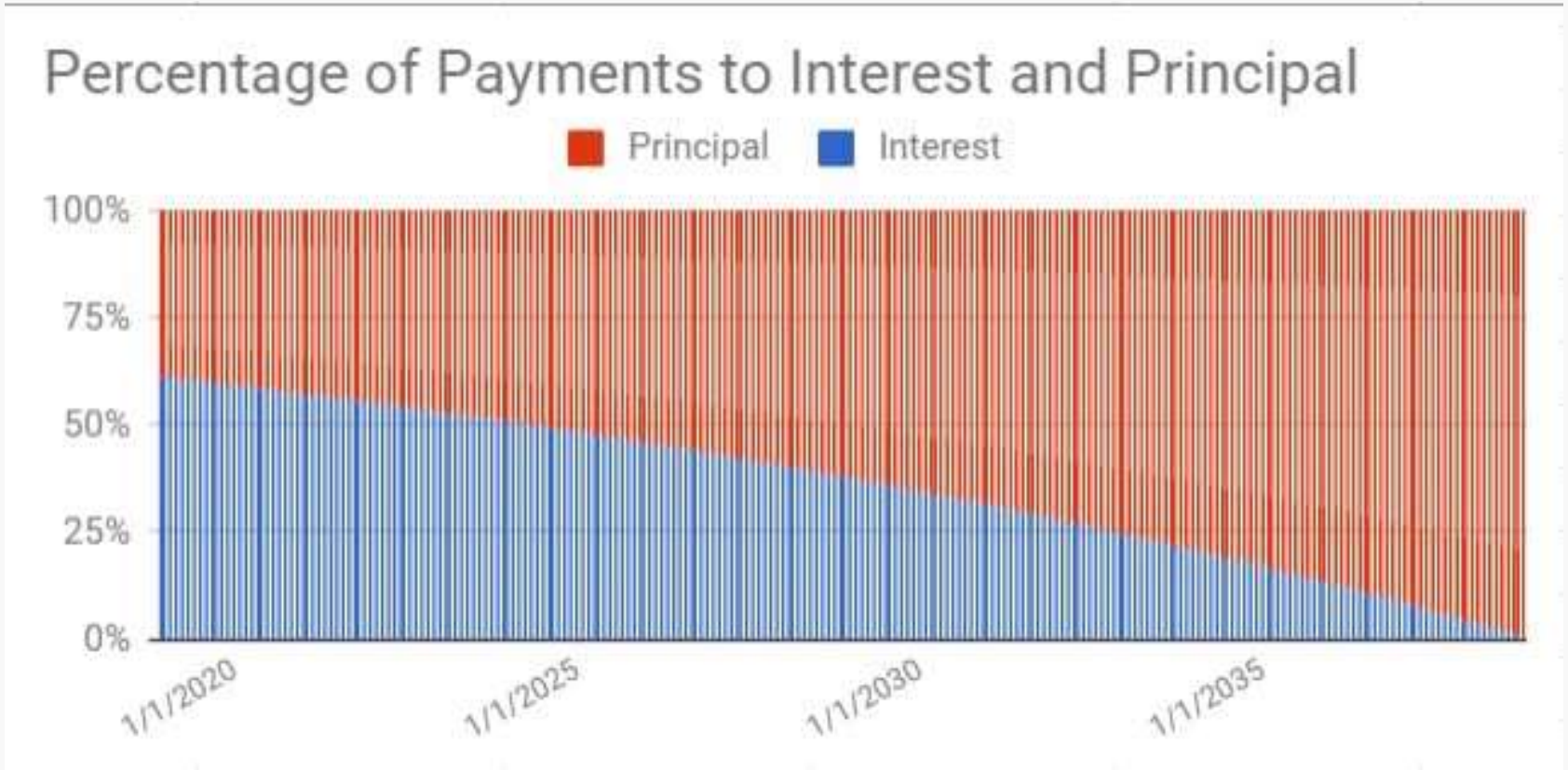
  4,0214
- Ending balance

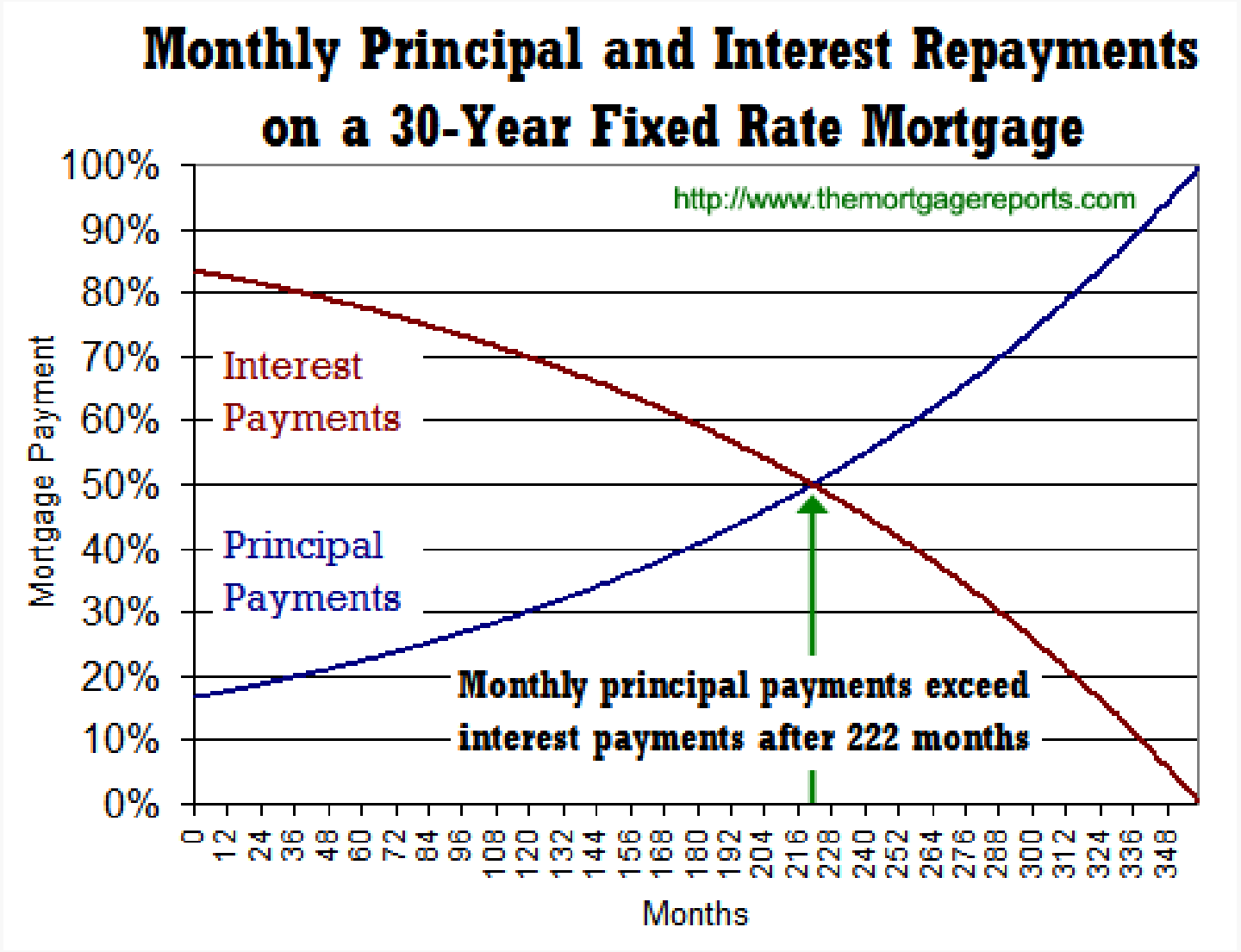
  −3,656−365+4,021=0
- The loan is fully paid off.

- OBSERVE THAT: 10% is only calculated over unpaid balance. These three annual payments of $4,021 repay the loan itself and provide a return of 10% on the amount still outstanding each year.



In general, the portion of interest in the yearly payments decline whereas the proportion of the principal increases




Monthly Principal and Interest Repayments
on a 30-Year Fixed Rate Mortgage
http://www.themortgagereports.com
Mortgage Payment
100%
90%
80%
70%
60%
50%
40%
30%
20%
10%
0%
Interest
Payments
Principal
Payments
Monthly principal payments exceed
interest payments after 222 months
0
12
24
36
48
60
72
84
96
108
120
132
144
156
168
180
192
204
216
228
240
252
264
276
288
300
312
324
336
348
Months

- If we calculate the *PW* of the loan transaction at this rate of return of 10%;

$$PW(10\%) = -\$10,000 + \$4,021(P/A, i, 3) = 0$$

- which indicates that 10% earned on the amount still outstanding each year by the bank is also the break even interest rate (largest MARR that can be met by this loan)

$$-10{,}000 + \frac{4,021}{(1+i)^1} + \frac{4{,}021}{(1+i)^2} + \frac{4{,}021}{(1+i)^3} = 0$$

We solve for i.

When $i = 10\%$, the equation becomes:

$$-10{,}000 + 3{,}655.\overline{45} + 3{,}323.\overline{13} + 3{,}021 = 0$$

Add the inflows:

$$3,655.45 + 3,323.13 + 3,021 = 10,000 \quad \text{(within rounding)}$$

So: $PW = 0$

- **Rate of Return: Definition 2**
- is the break-even interest i (−100% to 1) at which the PW of Project is zero.

$$PW(i^*) = PWCash\ Inflows - PWCash\ Outflows = 0$$

In detail, the equation $PW(i^*) = 0$ is equivalent to;

$$PW(i^*) = \frac{A_0}{(1+i^*)^0} + \frac{A_1}{(1+i^*)^1} + \cdots + \frac{A_N}{(1+i^*)^N} = 0$$

Here, we know all A_n values for each period, but not the value of i^*. Since it is the only unknown, we can solve for i^*. But the solution is not always straight forward and unique.

Note that by multiplying both sides of Equation (1) by $(1+i^*)^N$ and $(A/P, i^*, N)$ we obtain;

$$\begin{aligned} PW(i^*)(1+i^*)^N &= FW(i^*) = 0 \\ PW(i^*)(A/P, i^*, N) &= AW(i^*) = 0 \end{aligned}$$

- **Rate of Return: Definition 2 (Modified)**
- is the rate of interest i (−100% to 1), that equates the *PW*, *FW*, *AW* of the entire series of cash flows of a project to zero.

$$PW(i) = FW(i) = AW(i) = 0$$

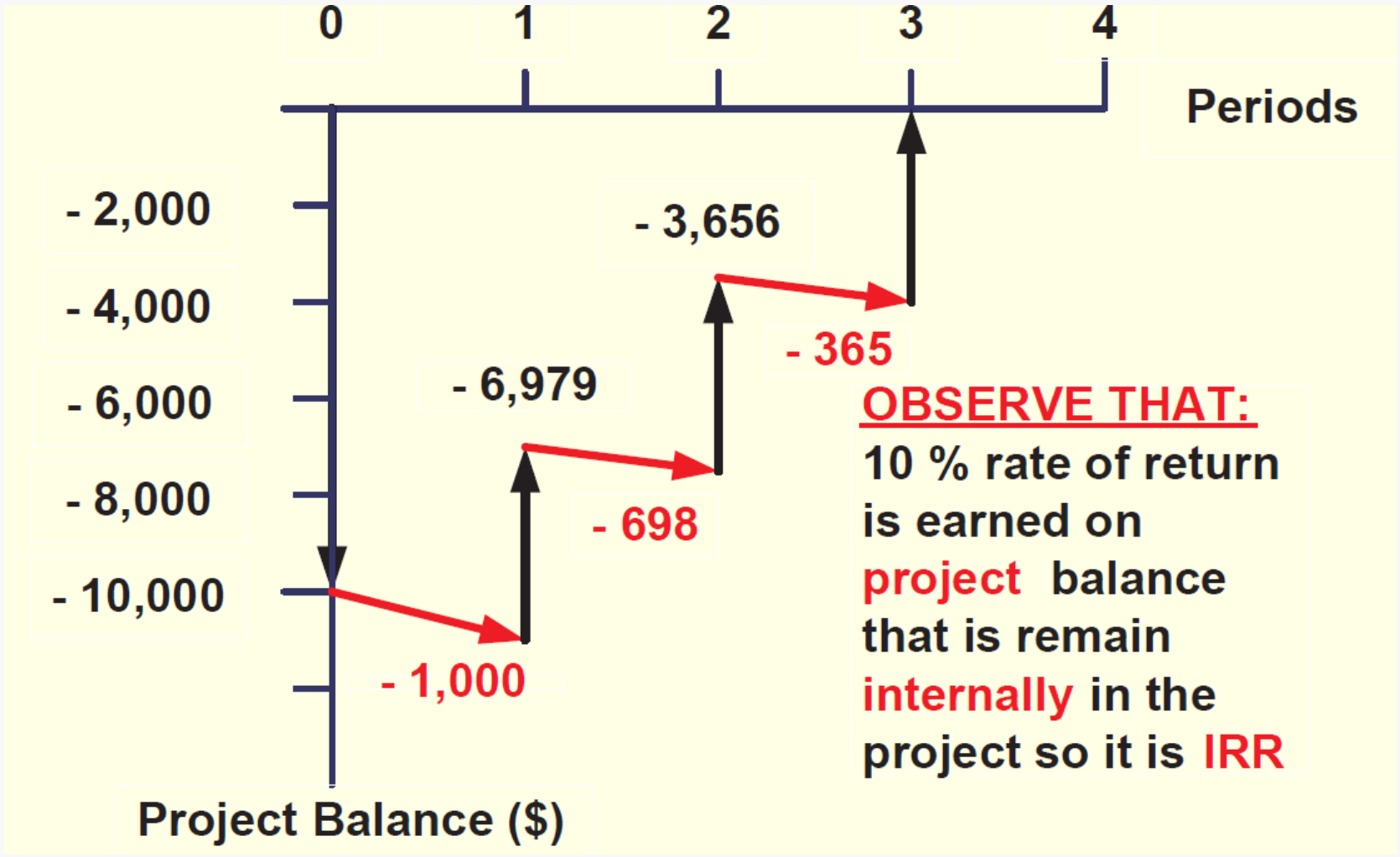
- Lastly, we will introduce the concept of rate of return based on the return on invested capital in terms of a project investment.
- A project's return is referred to as the internal rate of return (IRR) promised by an investment project over its useful life.

- **Internal Rate of Return: Definition 3**
- is the interest rate charged on the unrecovered project balance of the investment such that, when project terminates, the unrecovered Project balance is zero.

EXAMPLE

- Suppose a company invests $10,000 in a computer with a three-year useful life, which will create equivalent annual labor savings of $4,021 at the end of each year for three years. Here, we can view the investing firm as the lender and the project as the borrower.

Year	Beginning Project Balance	Return on Invested Capital	Cash Generated from Project	Project Balance at End
0				-$10,000
1	-$10,000	-$1,000	$4,021	-$6,979
2	-$6,979	-$698	$4,021	-$3,656
3	-$3,656	-$365	$4,021	$0



OBSERVE THAT:

10 % rate of return is earned on project balance that is remain internally in the project so it is IRR

Year	Beginning Project Balance	Return on Invested Capital	Cash Generated from Project	Project Balance at End
0				-$10,000
1	-$10,000	-$1,000	$4,021	-$6,979
2	-$6,979	-$698	$4,021	-$3,656
3	-$3,656	-$365	$4,021	$0

- Firm earns a 10% rate of returns on funds that remain internally invested in the project.
- Since it is a return internal to the project it is called IRR
- Computer project brings in enough cash to pay for itself in three years and to provide the firm a return of 10% on invested capital
- If the computer is financed with funds costing 10% annually, the cash generated by this investment will be exactly sufficient to repay the principal and its annual charge on the funds

- To facilitate the process of finding the rate of return for an investment project, we will first classify the types of investment cash flows.

Classification of Investments

We classify an investment by counting the number of sign changes in its net cash-flow sequence. A change from either "-" to "+" or "+" to "-" is counted as one sign change. (Ignore the zero cash flows!)

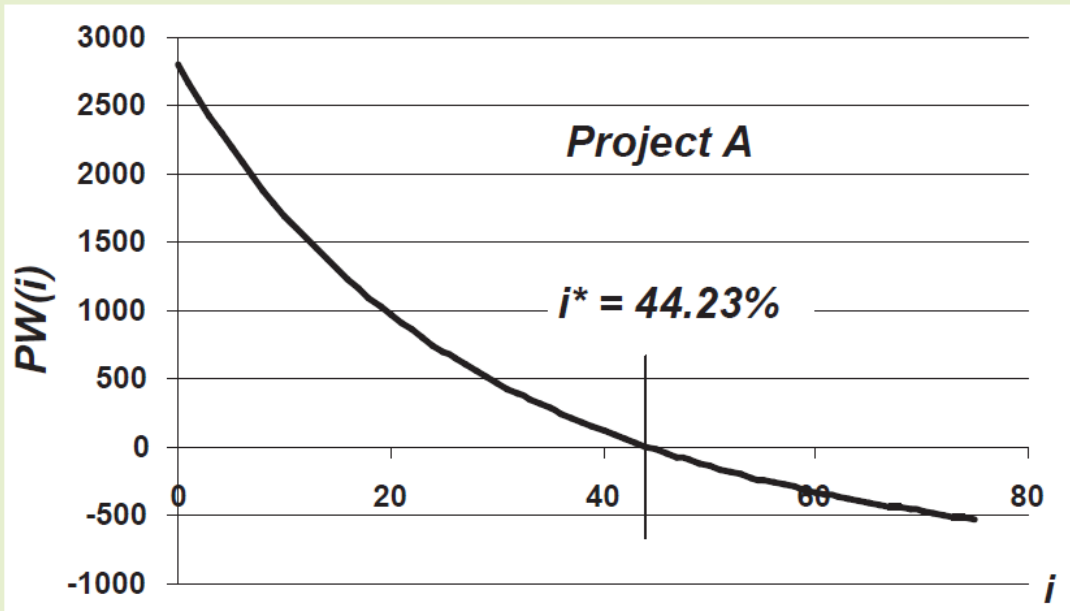
1. Simple (or Conventional) Investment
   - Initial cash flows are negative and only one sign change occurs
   - Unique solution exists
   - Question is: is it real or imaginary? If real, is it in $(-100, \infty)$ interval
   - A negative rate of return implies you never recover the initial investment
2. Simple Borrowing
   - Initial cash flows are positive and only one sign change occurs
   - Unique solution exists but this is not an investment flow
3. Nonsimple (or Nonconventional) Investment
   - more than one sign change occurs
   - Multiple i^* can occur

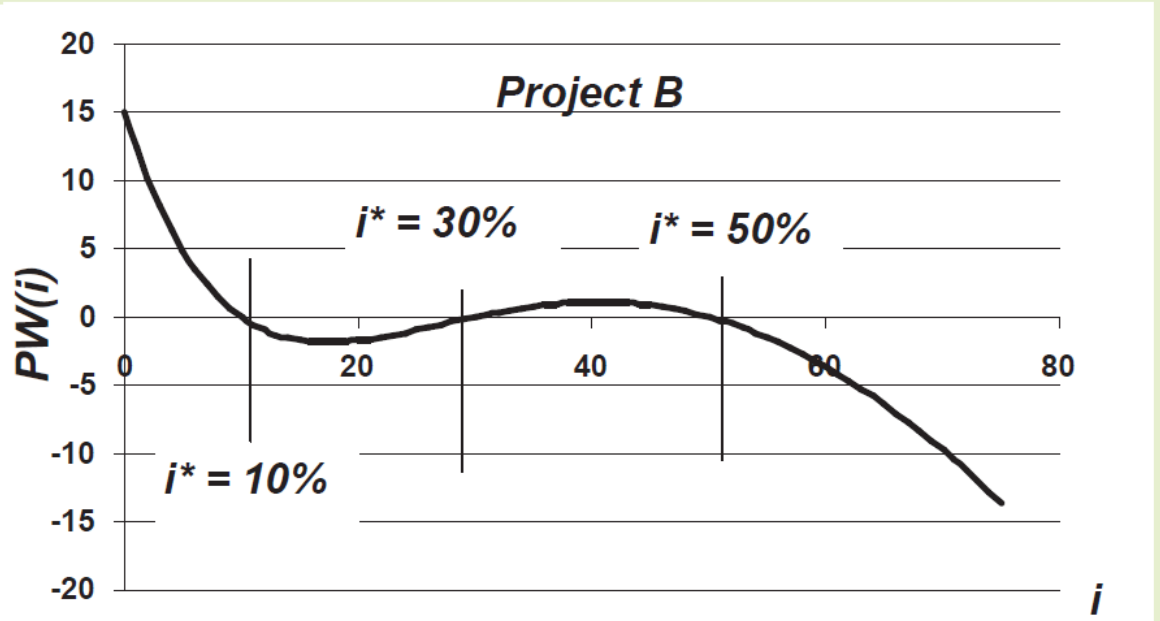
Example

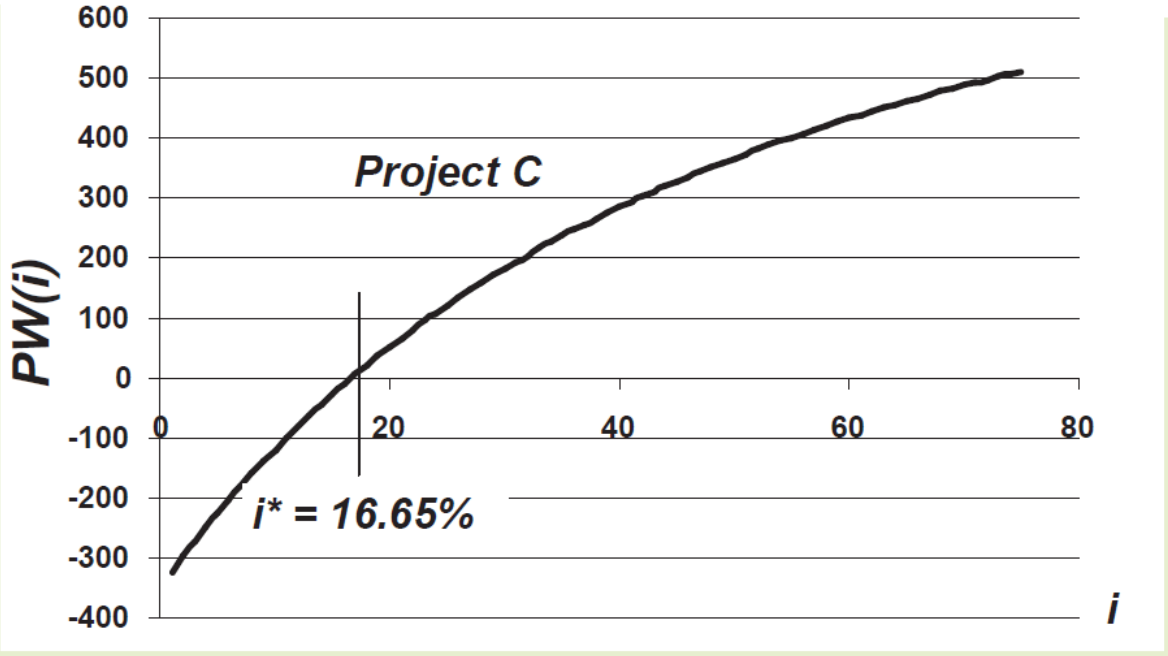
Classify the following three cash flow series;

Period *n*	Project A	Project B	Project C
0	-$1,000	-$1,000	$1,000
1	-$500	$3,900	-$450
2	$800	-$5,030	-$450
3	$1,500	$2,145	-$450
4	$2,000	0	0

- Project A is a simple investment (one sign change)
- Project B is a nonsimple investment (three sign change)
- Project C is a simple borrowing (one sign change)


Project A
PW(i)
i* = 44.23%
3000
2500
2000
1500
1000
500
0
-500
-1000
0
20
40
60
80
i


Project B
PW(i)
i* = 30%
i* = 50%
i* = 10%
20
15
10
5
0
-5
-10
-15
-20
0
20
40
60
80
i


Project C
PW(i)
i* = 16.65%
600
500
400
300
200
100
0
-100
-200
-300
-400
0
20
40
60
80
i

Internal rate of return

Let $(A_0, A_1, A_2, \ldots, A_n)$ be a cash flow stream. Then the internal rate of return of this stream is a number i^* satisfying the equation

$$0 = A_0 + \frac{A_1}{1+i^*} + \frac{A_2}{(1+i^*)^2} + \cdots + \frac{A_n}{(1+i^*)^n}.$$

Equivalently, it is a number i^* satisfying $x = \frac{1}{1+i^*}$, where x satisfies the polynomial equation

$$0 = A_0 + A_1 x + A_2 x^2 + \cdots + A_n x^n.$$

- IRR is defined without reference to the external financial World (external interest rate)
- It is determined entirely by the cash flow stream
- Equation for IRR is a polynomial equation in x of degree n
- Has at least one root, may have as many as n roots, but some orvall roots may be complex numbers

$$0 = A_0 + A_1 x + A_2 x^2 + \cdots + A_n x^n, \textbf{ where } x = \frac{1}{1 + i^*}$$

- A sign change in the cash flow (*Ai* 's) is where successive nonzero terms of *Ai* 's, written according to descending powers of *x*, have different signs.

Descartes' Rule

If a polynomial with real coefficients has *m* sign changes, then the number of positive roots will be $m - 2k$, where *k* is a positive integer or zero ($k = 0, 1, 2, 3, \ldots$).

number of sign changes *m*	Number of positive roots of *x*
0	0 (no IRR exists!)
1	1
2	2 or 0
3	3 or 1
4	4, 2, or 0

- But this rule is applicable to the roots, x.
- Since, $x = 1/(1+i)$, a positive root, x does not ensure a positive value for i^*.
- There may be as many positive values of i (root of ROR equation) as there are sign changes in the cash flow.

Main Theorem of Internal rate of return

Suppose the cash flow stream $(A_0, A_1, A_2, \ldots, A_n)$ has $A_0 < 0$ and $A_k \geq 0$ for all $k, k = 1, 2, \ldots, n$ with at least one term strictly positive. Then there is a unique positive root to the equation

$$0 = A_0 + A_1 x + A_2 x^2 + \cdots + A_n x^n.$$

Furthermore, if $\sum_{k=0}^{n} A_k > 0$ (meaning that the total amount returned exceeds the initial investment), then the compounding internal rate of return $i = (1/x) - 1$ is positive.

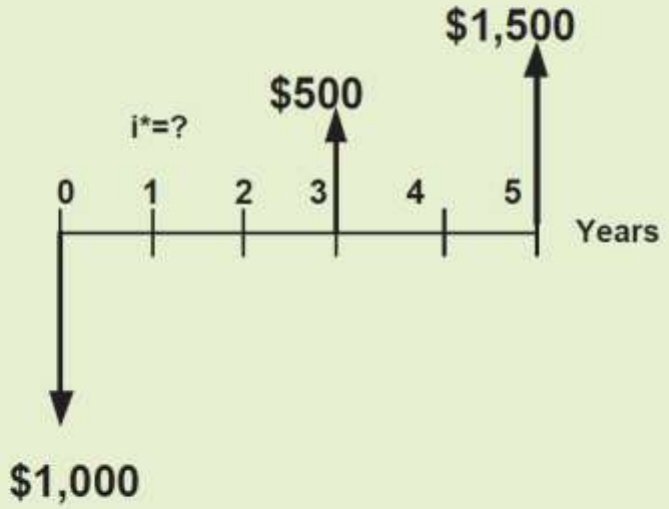
- The conditions in the theorem ensure a single positive *i* root, for one sign change.
- If there is more than one sign change in the net cash flows, it is possible that there will be multiple *i* values.

- To determine the ROR of a cash flow series, set up the ROR equation using either *PW* or *AW* relations.
- For *PW*:
  - Let the present worth of costs or disbursements to be PW_D
  - Let the present worth of incomes or receipts to be PW_R
  - Solve $0 = PW_R - PW_D$ for i
- Similarly for *AW*:
  - Solve $0 = AW_R - AW_D$ for i
- The i value that makes these values numerically correct is called i^*. It is the root of the ROR relation.
- Compare i^* with MARR to decide if the cash flow series is viable:
  - If $i^* \geq MARR$, accept the alternative
  - If $i^* < MARR$, reject the alternative
- But this i^* must be internal to the project to be an IRR, and we need to check this even if it is a unique positive rate.

- In the rate of return calculations, objective is to find the interest rate i^* at which the positive and negative cash flows are equivalent.

Example

If you deposit \$1,000 now and promised payments of \$500 three years from now and \$1,500 five years from now, find the rate of return using *PW* factors?



$$0 = -1,000 + 500(P/F, i\%, 3) + 1500(P/F, i\%, 5) \rightarrow i^* = 16,9\%$$

By Trial and Error.

Trial and Error

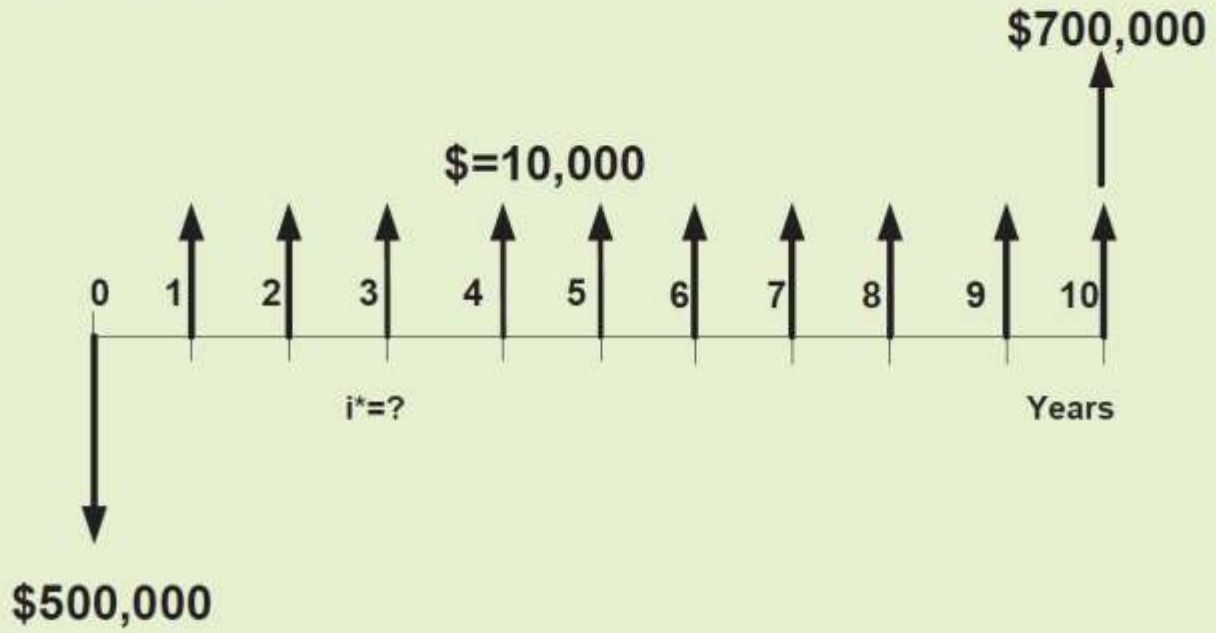
- Draw a cash flow diagram
- Set up the rate of return equation
- Select the i values until the equation is balanced

Choosing the first i as closer to i^* is important. The procedure for obtaining a good initial i is:

1. Convert all disbursements into either single amounts or uniform values while neglecting the time value of money
2. Convert all receipts into either single amounts or uniform values while neglecting the time value of money
3. Combine the receipts and disbursements so that a P/F, P/A, or A/F format applies
4. Use interest rate tables to find the approximate interest rate at which P/F, P/A, or A/F value is satisfied. The obtained rate is a good initial rate for trial and error.

Example

A construction company has requested that $500,000 be spent now during construction on software and hardware to improve the efficiency of environmental control systems for the construction of a skyscraper. This is expected to save $10,000 per year for 10 years in energy costs and $700,000 at the end of 10 years in equipment refurbishment costs. Find the rate of return?



1. Cash flow diagram
2. $0 = -500,000 + 10,000(P/A, i\%, 10) + 700,000(P/F, i\%, 10)$
3. Take all income as a single Future amount:

$$500,000 = 800,000(P/F, i, 10) \rightarrow (P/F, i, 10) = 0.625$$

The rate i is between 4% and 5%. Use 5% as the initial one.

$$6,946 = -500,000 + 10,000(P/A, 5\%, 10) + 700,000(P/F, 5\%, 10)$$

indicating the return is more than 5%. Try 6%:

$$\begin{aligned} -35,519 &= -500,000 + 10,000(P/A, 6\%, 10) \\ &+ 700,000(P/F, 6\%, 10) \end{aligned}$$

4. Interpolate between 5% and 6%:

$$i^* = 5 + \frac{6946 - 0}{6946 - (-35,519)}(1.0) = 5.16\%$$

- Linear interpolation estimates a value that lies between two known values by assuming the change between them is linear (a straight line).
- Interpolation formula:

For two points:

$$(i_1, PW_1) = (5\%, +6946)$$

$$(i_2, PW_2) = (6\%, -35519)$$

Rate where PW = 0:

$$i^* = i_1 + \frac{PW_1 - 0}{PW_1 - PW_2}(i_2 - i_1)$$

- Calculation:

$$i^* = 5 + \frac{6946}{6946 - (-35519)}(1)$$

Now compute the denominator:

$$6946 - (-35519) = 6946 + 35519 = 42465$$

So:

$$i^* = 5 + \frac{6946}{42465}$$

$$\frac{6946}{42465} = 0.1636$$

$$i^* = 5 + 0.1636 = 5.1636\%$$

$$i^* \approx 5.16\%$$

When applied correctly, the ROR technique will always result in a correct decision, indeed, the same one as with a *PW* or *AW* analysis.

- However, there are some assumptions and difficulties with ROR analysis while calculating i^* and interpreting its real-world meaning.
  1. Depending on the sequence of net cash flows, there may be more than one real-number root to the ROR equation, resulting in more than one i^* value.
  2. ROR method assumes that any net positive investment are reinvested at rate i^*
  3. Obtaining one or multiple i^* values by trial-and-error solution is not easy
  4. A unique approach must be applied to choose from two or more mutually exclusive alternatives

In Class Work 9

During the last 3 years, the net cash flows for contract payments have varied widely, as shown below:

Year	0	1	2	3
Cash Flow	2,000	-500	-8,100	6,800

1. Determine the maximum number of i^* values that may satisfy the ROR relation.
2. Write the PW-based ROR relation and approximate the i^* value(s), using 0, 5, 10, 20, 30, 40, and 50%.

1. There are two sign changes, therefore it indicates a maximum of two i^* values.
2.

i%	0	5	10	20	30	40	50
PW							

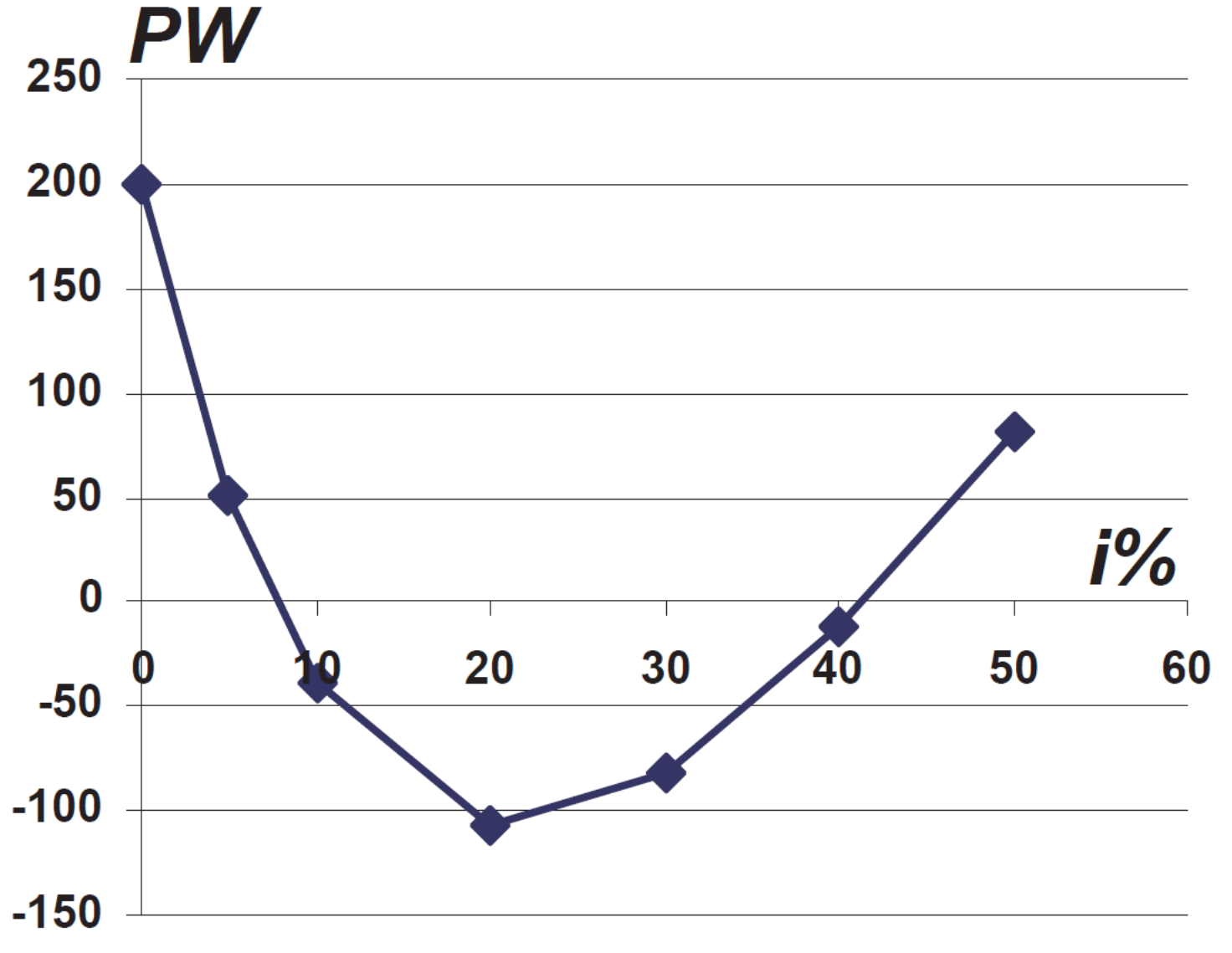
1. Two sign changes, maximum of two i^* values.
2.

$$\begin{aligned} PW &= 2,000 - 500(P/F, i\%, 1) - 8,100(P/F, i\%, 2) \\ &+ 6,800(P/F, i\%, 3) \\ &= 2,000 - \frac{500}{1+i} - \frac{8,100}{(1+i)^2} + \frac{6,800}{(1+i)^3} \end{aligned}$$

i%	0	5	10	20	30	40	50
PW	200	50.97	-39.82	-106.48	-82.38	-11.66	81.48

$$i_1^* = 5 + \frac{0 - 50.97}{-39.82 - 50.97}(5.0) = 7.81\%$$

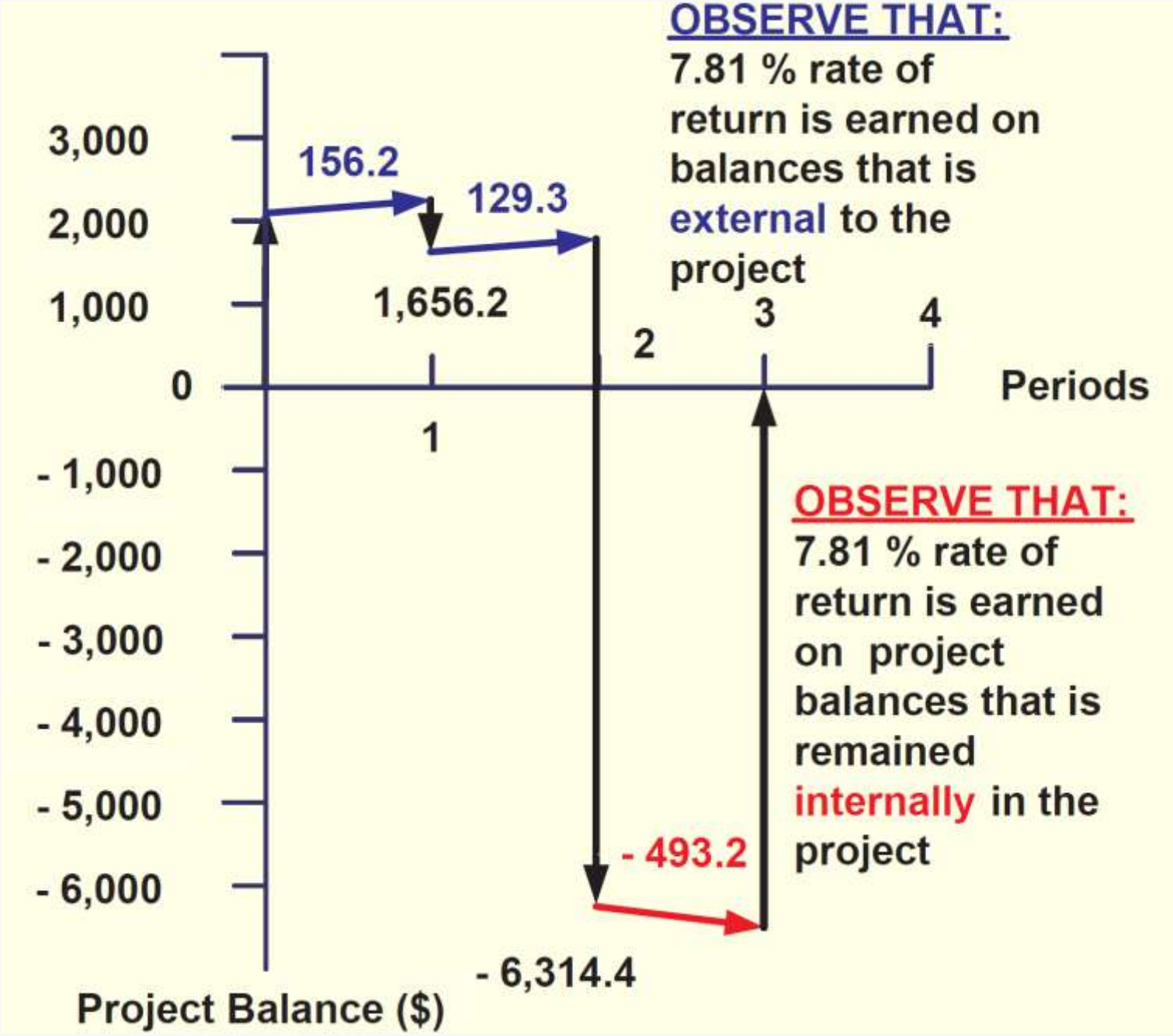
$$i_2^* = 40 + \frac{0 - (-11.66)}{81.48 - (-11.66)}(10) = 41.25\%$$


PW
250
200
150
100
50
0
-50
-100
-150
0
10
20
30
40
50
60
i%

Are these rates IRR?

By definition; IRR is the rate charged on the unrecovered project balance of the investment.

- So the rate is applicable to the internal balances not external


OBSERVE THAT:
7.81 % rate of return is earned on balances that is external to the project
3,000
2,000
1,000
0
- 1,000
- 2,000
- 3,000
- 4,000
- 5,000
- 6,000
156.2
129.3
1,656.2
- 493.2
- 6,314.4
1
2
3
4
Periods
OBSERVE THAT:
7.81 % rate of return is earned on project balances that is remained internally in the project
Project Balance ($)

ACTIVITY

- Your computer engineering lab needs new laptops. Three models are available.
- Use Annual Worth (AW) analysis to determine which laptop is the most economical option.
- Assume: MARR = 10%.

- Laptop Options

Laptop A — Basic Dev Machine

- Purchase price: $1,200
- Annual maintenance: $150
- Battery replacement: $40 per year
- Life: 3 years
- Salvage value: $150

Laptop B — Performance Model

- Purchase price: $1,600
- Annual maintenance: $120
- Battery replacement: $80 every 2 years
- Life: 4 years
- Salvage value: $250

Laptop C — High-End Engineering Laptop

- Purchase price: $2,200
- Annual maintenance: $90
- Battery replacement: $50 per year
- Life: 5 years
- Salvage value: $300

Instructions

Step 1. Compute capital recovery (CR):

$$CR = -P(A/P, 10\%, n) + S(A/F, 10\%, n)$$

Step 2. Add annual operating costs:

$$AW = CR + (\text{maintenance}) + (\text{battery cost})$$

Step 3. Compare AW values and choose the laptop with the lowest AW.

$$\boxed{A/P = \frac{i(1+i)^n}{(1+i)^n - 1}}$$

$$\boxed{A/F = \frac{i}{(1+i)^n - 1}}$$

ACTIVITY 2

- The Computer Engineering lab is considering whether to keep the existing PCs or replace them with new, energy-efficient machines. You will perform an incremental analysis to evaluate the upgrade.
- Assume: All cash flows are in $ per year. Study period is 5 years (the life of the new PCs).

Option 1: Keep Old PCs

- No initial investment (the old PCs are already owned).
- Annual maintenance cost: \$4,000 per year.
- Old PCs are less efficient and cause extra electricity cost, but no additional salvage value is considered.

Option 2: Buy New PCs

- Initial cost of new PCs: \$30,000 at year 0.
- Annual maintenance cost: \$1,000 per year.
- Annual energy savings compared to old PCs: \$1,000 per year.
- Salvage value at the end of year 5: \$5,000.
- Useful life: 5 years.

Tasks

(a) Construct the incremental cash flow (Upgrade – Keep Old).
Fill in the incremental cash flow table (from the point of view of "Upgrade" vs. "Keep Old"):

Year	Incremental Cash Flow (Upgrade – Keep Old)
0	_____
1	_____
2	_____
3	_____
4	_____
5	_____

(b) Estimate the IRR of the upgrade using the incremental cash flow.
Is the internal rate of return (IRR) positive or negative? Based on the sign and magnitude of IRR, would you recommend upgrading?

IRR $\approx$ _____

(c) Use Annual Worth (AW) to check your decision.
Assume MARR (minimum attractive rate of return) is 10%. Compute the AW of the incremental cash flow at 10%:

$$AW_\Delta = \underline{\qquad\qquad\qquad\qquad} \quad \text{(upgrade vs. keep old)}$$

If $AW_\Delta > 0$, the upgrade is economically justified. If $AW_\Delta < 0$, keeping the old PCs is preferred.